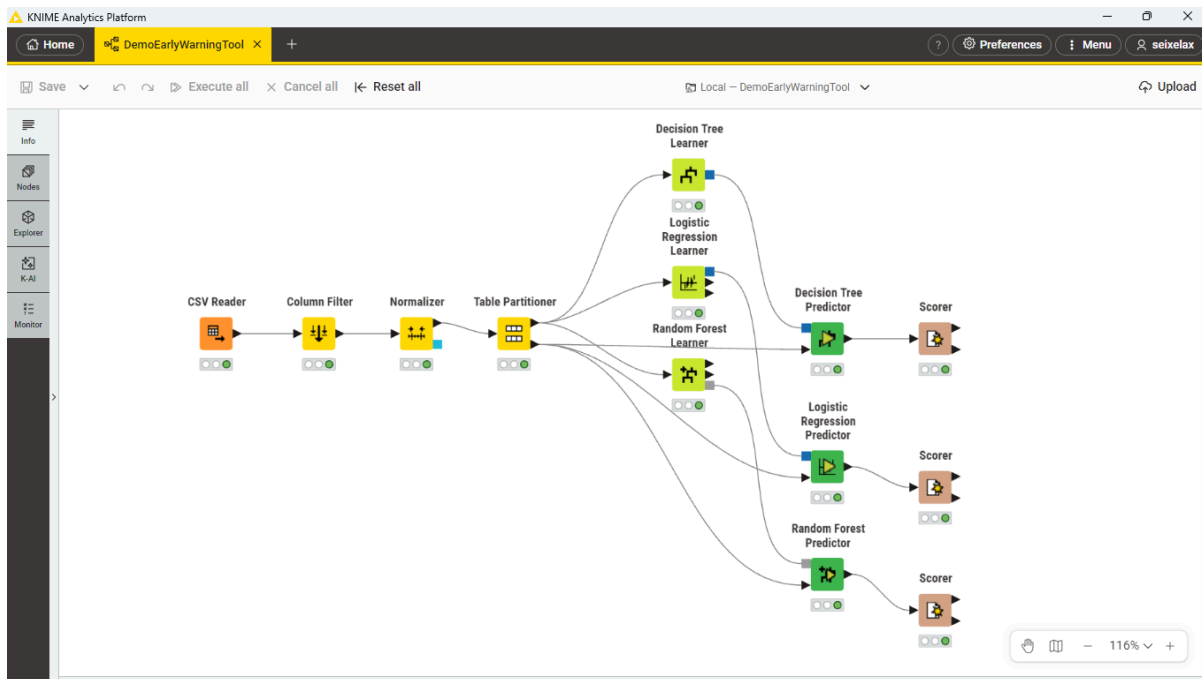


Complete KNIME Workflow for Student Early Warning Tool



CSV Reader Node and Dataset Preview

#	RowID	student_id	attendance	quiz_score	assignment_score	exam_score	risk_status
1	Row0	S001	95	90	92	88	Not At Risk
2	Row1	S002	88	84	86	82	Not At Risk
3	Row2	S003	76	72	75	70	Not At Risk

Decision Tree

KNIME Analytics Platform

Home DemoEarlyWarningTool + Preferences Menu seixelax

Save Execute Cancel Reset Local - DemoEarlyWarningTool Upload

Scorer

First column: Prediction (risk_status)

Second column: risk_status

Sorting strategy: Insertion order

Missing values: Ignore Fail

Discard Apply Apply and Execute

► 1: Confusion matrix ► 2: Accuracy statistics Flow Variables

#	RowID	Not At Risk	At Risk
1	Not A...	3	0
2	At Risk	0	3

Logistic Regression

KNIME Analytics Platform

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Scorer

First column: Prediction (risk_status)

Second column: risk_status

Sorting strategy: Insertion order

Missing values: Ignore Fail

Discard Apply Apply and Execute

► 1: Confusion matrix ► 2: Accuracy statistics Flow Variables

#	RowID	Not At Risk	At Risk
1	Not A...	3	2
2	At Risk	0	1

Random Forest

KNIME Analytics Platform

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Preferences Menu seixelax

Save Execute Cancel Reset

Local - DemoEarlyWarningTool Upload

Info Nodes Explorer K-AI Monitor

```
graph LR; CSVReader[CSV Reader] --> ColumnFilter[Column Filter]; ColumnFilter --> Normalizer[Normalizer]; Normalizer --> TablePartitioner[Table Partitioner]; TablePartitioner --> DTL[Decision Tree Learner]; TablePartitioner --> LRL[Logistic Regression Learner]; TablePartitioner --> RFL[Random Forest Learner]; DTL --> DTP[Decision Tree Predictor]; LRL --> LRP[Logistic Regression Predictor]; RFL --> RFP[Random Forest Predictor]; DTP --> S1[Scorer]; LRP --> S2[Scorer]; RFP --> S3[Scorer]; S1 --> S4[Scorer]; S2 --> S4; S3 --> S4; S4 --> S5[Scorer];
```

Scorer

First column
Prediction (risk_status)

Second column
risk_status

Sorting strategy
Insertion order

☐ Reverse order

☐ Use name prefix

Missing values
Ignore Fail

Discard Apply Apply and Execute

► 1: Confusion matrix ► 2: Accuracy statistics

Flow Variables

Rows: 2 | Columns: 2

Table Statistics

Search

	#	RowID	Not At Risk (#) Number (Integer)	At Risk (#) Number (Integer)
☐	1	Not A...	3	0
☐	2	At Risk	0	3